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Factors, primes and divisibility

Level 2 Intermediate · Number theory

How to read divisors, HCFs, LCMs and factorials straight off a prime factorisation.

By the end you can: count the divisors of a number, and the divisors with a given property, from its prime factorisation; find HCFs and LCMs by comparing exponents or by Euclid's subtraction idea; find the exact power of a prime that divides a factorial.

Lesson

How many positive whole numbers divide 3600? You could list pairs: 1 and 3600, 2 and 1800, 3 and 1200, and so on, but by the twentieth pair you would be worrying about whether you had missed one. Here is the better way. Write $3600 = 2^4 \times 3^2 \times 5^2$. Every divisor of 3600 is built by choosing how many 2s, how many 3s and how many 5s to take: between 0 and 4 twos, between 0 and 2 threes, and between 0 and 2 fives. That is $5 \times 3 \times 3 = 45$ choices, so 3600 has exactly 45 divisors. No listing, no worrying.

This unit is about that one move: break a number into primes, then read off what you want from the exponents. Divisors, highest common factors, lowest common multiples, perfect squares and factorials all turn into bookkeeping with exponents.

Primes and prime factorisation

A prime is a positive integer with exactly two positive divisors, 1 and itself. So 1 is not prime (it has only one divisor), and 2 is the only even prime. The first few primes are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

> **The Fundamental Theorem of Arithmetic.** Every integer $n > 1$ can be written as a product of primes, and in only one way apart from the order of the factors.

Existence is easy. If n is prime, stop. Otherwise $n = ab$ with $1 < a, b < n$, so split a and b in the same way. The numbers keep getting smaller, so the process must end. Uniqueness rests on one fact, *Euclid's lemma*: if a prime p divides a product ab, then p divides a or p divides b. (We take it on trust here; you will see why it is true when we meet modular inverses in Level 3.) Given two factorisations of the same number, each prime in the first divides the product in the second, so by the lemma it is one of the primes there. Cancel it from both sides and repeat.

To test whether n is prime you only need to try primes up to $\sqrt{n}$. If $n = ab$ with $a \le b$, then $a^2 \le ab = n$, so $a \le \sqrt{n}$: any factorisation has a factor no bigger than the square root. To check 211, try 2, 3, 5, 7, 11 and 13. None divides it, and $17^2 = 289 > 211$, so 211 is prime.

Setters love the current year, so know its factorisation by heart: $2024 = 2^3 \times 11 \times 23$, $2025 = 3^4 \times 5^2 = 45^2$, $2026 = 2 \times 1013$ with 1013 prime, and 2027 is prime.

Reading a number from its factorisation

Write $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$. A positive integer d divides n exactly when $d = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}$ with $0 \le b_i \le a_i$ for every i. In words: a divisor uses only the primes of n, and never more copies of a prime than n has. (If d involved a prime q not in the list, then q would divide n, which uniqueness forbids.)

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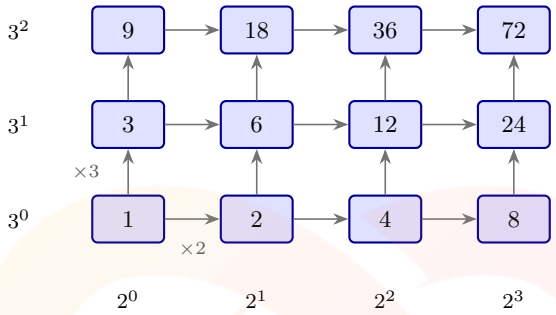
Counting and adding divisors. If $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, then the number of positive divisors of n is

$$d(n) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1),$$

and the sum of the positive divisors is

$$\sigma(n) = (1 + p_1 + \cdots + p_1^{a_1})(1 + p_2 + \cdots + p_2^{a_2}) \cdots (1 + p_k + \cdots + p_k^{a_k}).$$

The count is just the multiplication principle: there are $a_1 + 1$ choices for the exponent of p_1 (remember that 0 is a choice), $a_2 + 1$ for the exponent of p_2, and so on. The sum works because expanding the brackets produces one term for each choice of exponents, so each divisor appears once.



4 choices for the power of 2, 3 for the power of 3: $4 \times 3 = 12$ divisors

The picture shows this for $72 = 2^3 \times 3^2$. Put the divisors in a grid with the power of 2 going across and the power of 3 going up. There are 4 columns and 3 rows, so $d(72) = 12$.

Divisors also come in pairs: d goes with n/d. In each pair one member is below $\sqrt{n}$ and the other is above, except when $d = \sqrt{n}$, which pairs with itself. So n **has an odd number of divisors if and only if** n **is a perfect square**. The formula agrees: $(a_1 + 1) \cdots (a_k + 1)$ is odd only when every a_i is even.

The real power is in counting divisors *with a property*. You simply restrict the exponent choices. A divisor is odd when its exponent of 2 is 0. It is a multiple of 10 when its exponents of 2 and 5 are both at least 1. It is a perfect square when every exponent is even.

Example 1. How many positive divisors of 5400 are multiples of 15?

Solution.
Listing the multiples of 15 up to 5400 and testing each one would take all afternoon. Factorise instead: $5400 = 54 \times 100 = (2 \times 3^3) \times (2^2 \times 5^2) = 2^3 \times 3^3 \times 5^2$. A divisor has the form $2^a 3^b 5^c$ with $0 \le a \le 3$, $0 \le b \le 3$ and $0 \le c \le 2$.
Being a multiple of $15 = 3 \times 5$ means $b \ge 1$ and $c \ge 1$. So a has 4 choices, b has 3 (namely 1, 2 or 3) and c has 2, giving $4 \times 3 \times 2 = 24$ divisors.
There is a second way to see it, which makes a good check. The divisors we want are the numbers $15k$ that divide 5400, which happens when k divides $5400/15 = 360$. Since $360 = 2^3 \times 3^2 \times 5$ has $4 \times 3 \times 2 = 24$ divisors, the two answers agree.

Example 2. What is the smallest positive integer with exactly 12 positive divisors?

Solution.
Testing numbers one at a time does work eventually, but it teaches you nothing, and the next question will ask for 48 divisors. Turn it round. We need $(a_1 + 1)(a_2 + 1) \cdots = 12$, so first list the ways to write 12 as a product of factors that are each at least 2: 12, 6×2, 4×3 and

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$3 \times 2 \times 2$. These give the exponent patterns (11), (5, 1), (3, 2) and (2, 1, 1).
For each pattern the smallest number puts the biggest exponent on the smallest prime. That gives $2^{11} = 2048$, $2^5 \times 3 = 96$, $2^3 \times 3^2 = 72$ and $2^2 \times 3 \times 5 = 60$. The smallest is 60.
Spreading the exponents over several small primes usually wins, since piling up powers of 2 doubles the number again and again. But do not trust that rule blindly: always compare the candidates.

HCF and LCM

HCF and LCM from factorisations. If $a = \prod p^{\alpha_p}$ and $b = \prod p^{\beta_p}$ (allowing exponent 0), then

$$\mathrm{hcf}(a,b) = \prod p^{\min(\alpha_p,\beta_p)}, \qquad \mathrm{lcm}(a,b) = \prod p^{\max(\alpha_p,\beta_p)}.$$

Since $\min(x,y) + \max(x,y) = x + y$, it follows that $\mathrm{hcf}(a,b) \times \mathrm{lcm}(a,b) = ab$.

These come straight from the previous section. A common divisor may use each prime no more often than *both* numbers do, and a common multiple must use each prime at least as often as *either* does. For example, $360 = 2^3 \times 3^2 \times 5$ and $84 = 2^2 \times 3 \times 7$, so $\mathrm{hcf}(360, 84) = 2^2 \times 3 = 12$ and $\mathrm{lcm}(360, 84) = 2^3 \times 3^2 \times 5 \times 7 = 2520$. Check: $12 \times 2520 = 30240 = 360 \times 84$.

When the numbers are large and awkward, factorising can be slow. Then use Euclid's idea: **any common divisor of a and b also divides $a - kb$ for every integer k**, and any common divisor of $a - kb$ and b divides a. So $\mathrm{hcf}(a,b) = \mathrm{hcf}(a - kb, b)$, and you can keep replacing the bigger number by its remainder on division by the smaller:

$$\mathrm{hcf}(1001, 221) = \mathrm{hcf}(117, 221) = \mathrm{hcf}(117, 104) = \mathrm{hcf}(13, 104) = 13.$$

The same idea handles remainder questions. If dividing a by n leaves remainder r, then n divides $a - r$. So if two numbers leave known remainders on division by the same n, subtract the remainders and look for common divisors of what is left.

Example 3. How many ordered pairs (a, b) of positive integers have $\mathrm{hcf}(a,b) = 12$ and $\mathrm{lcm}(a,b) = 360$?

Solution.
A natural first idea is to use $ab = 12 \times 360 = 4320$ and count the factor pairs of 4320. That overcounts badly: $48 \times 90 = 4320$, but $\mathrm{hcf}(48, 90) = 6$. The product alone does not pin down the HCF.
The clean method is to pull the HCF out. Write $a = 12x$ and $b = 12y$. Then x and y have no common factor bigger than 1, since otherwise the HCF of a and b would be bigger than 12. For two such numbers the LCM is just the product, so $\mathrm{lcm}(a,b) = 12xy$, and the condition becomes $xy = 30$.
So we need ordered pairs of positive integers with product $30 = 2 \times 3 \times 5$ and no common factor. That means each prime goes entirely to x or entirely to y. Each of the three primes has 2 choices, so there are $2^3 = 8$ ordered pairs. They are $(a,b) = (12, 360), (24, 180), (36, 120), (60, 72)$ and their reverses, and the answer is 8.

Primes inside factorials

Factorials turn up constantly, and you need their factorisations without multiplying them out. How many times does 2 divide 12!? Every even number up to 12 gives one factor of 2, which is $\lfloor 12/2 \rfloor = 6$ factors. The multiples of 4 each give a second one, which is $\lfloor 12/4 \rfloor = 3$ more. The multiples of 8 give a third, which is 1 more, and there are no multiples of 16. The total is $6 + 3 + 1 = 10$.

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Legendre's formula. The exponent of the prime p in $n!$ is

$$\left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots,$$

where the terms become 0 once $p^k > n$.

Here $\lfloor x \rfloor$ means x rounded down. A number divisible by p^j but not by p^{j+1} is counted once in each of the first j terms, so j times in all, which is exactly how many factors of p it contributes. A famous special case: the number of zeros at the end of $n!$ is the exponent of 5, because there are always more 2s than 5s. So 100! ends in $20 + 4 = 24$ zeros.

Example 4. How many positive divisors of 12! are perfect squares, and what is the largest of them?

Solution.
Before counting anything we need 12! as a product of primes. Multiplying it out to get 479001600 and then factorising would be silly. Legendre's formula gives the exponents directly: for 2 it is $6 + 3 + 1 = 10$, for 3 it is $4 + 1 = 5$, for 5 it is 2, and for 7 and 11 it is 1. So $12! = 2^{10} \times 3^5 \times 5^2 \times 7 \times 11$.
A square divisor needs every exponent even. The exponent of 2 can be 0, 2, 4, 6, 8 or 10 (6 choices), the exponent of 3 can be 0, 2 or 4 (3 choices), the exponent of 5 can be 0 or 2 (2 choices), and the exponents of 7 and 11 must be 0. That gives $6 \times 3 \times 2 = 36$ square divisors.
The largest takes the biggest even exponent each time: $2^{10} \times 3^4 \times 5^2 = (2^5 \times 3^2 \times 5)^2 = 1440^2$.
A good habit: once you have a count like this, test the method on a tiny case. It predicts that $12 = 2^2 \times 3$ has $2 \times 1 = 2$ square divisors, and indeed they are 1 and 4.

Common traps

- 1 is not prime, but it divides everything. If a question says "factors other than 1 and the number itself", subtract 2 from the count.
- The number of divisors uses $a_i + 1$, not a_i. Forgetting that 0 is an allowed exponent is the commonest slip of all.
- Divisible by 4 and by 6 means divisible by $\text{lcm}(4, 6) = 12$, not by 24. You may multiply the two numbers only when they have no common factor.
- $\text{hcf} \times \text{lcm} = ab$ is true for two numbers, not three: $\text{hcf}(2, 2, 2) \times \text{lcm}(2, 2, 2) = 4$, not 8.
- Knowing ab does not tell you $\text{hcf}(a, b)$. Pull out the HCF first, as in Example 3.

How to spot this in a contest

The words *factor*, *divisor*, *multiple*, *prime* and *perfect square* are the obvious signals, but the idea also hides inside stories. Things that repeat and coincide (bells, buses, flashing lights, meshing gears) are LCM questions. Cutting a rectangle into equal squares, or sharing things into equal groups with nothing left over, are HCF questions. "The product is a perfect square" means check whether every exponent is even, and "a perfect cube" means every exponent is a multiple of 3. Trailing zeros, or "the largest power of p that divides", mean Legendre. Remainders from the same divisor mean subtract and take a common factor.

Above all, whenever a large specific number appears, especially the year, your first move should be to factorise it. If you find yourself listing more than about ten things, stop and ask whether the exponents could do the counting for you.

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Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ Call a positive integer *thrifty* if it has exactly three positive divisors. What is the sum of all thrifty numbers less than 200?

2. ⋆⋆ How many positive divisors of 7200 are multiples of 6 but not multiples of 12?

3. ⋆⋆ For how many positive integers n is $\text{lcm}(12, 18, n) = 720$?

4. ⋆⋆⋆ As n runs through the positive integers, what is the largest possible value of $\text{hcf}(n^2+5,\ n+2)$?

5. ⋆⋆⋆ A positive integer n has exactly 12 positive divisors. The number $2n$ has exactly 15 positive divisors, and $3n$ has exactly 16. Find n.

6. ⋆⋆⋆⋆ For how many positive integers $n \leq 1000$ does n have an odd number of odd positive divisors?

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. UKMT IMC 2016 Question 16 ⋆
2. AMC 8 2022 Problem 3 ⋆
3. UKMT IMC 2021 Question 6 ⋆
4. AMC 8 2018 Problem 18 ⋆⋆
5. AMC 10A 2022 Problem 7 ⋆⋆
6. AMC 10A 2023 Problem 5 ⋆⋆
7. UKMT IMC 2019 Question 24 ⋆⋆⋆
8. AMC 10A 2019 Problem 11 ⋆⋆⋆
9. AMC 10B 2017 Problem 20 ⋆⋆⋆
10. AMC 10B 2021 Problem 12 ⋆⋆⋆
11. UKMT IMC 2023 Question 23 ⋆⋆⋆
12. AMC 10A 2018 Problem 22 ⋆⋆⋆⋆

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Hints

1. Use the divisor-counting formula. Which exponent patterns give a product of brackets equal to 3?
2. Factorise 7200, then work out what being a multiple of 6 but not of 12 says about the exponent of 2 and the exponent of 3.
3. Work one prime at a time. For each prime, the largest of the three exponents must match the exponent in 720.
4. A common divisor of two numbers divides any combination of them. Which multiple of $n+2$ is very close to n^2+5?
5. Write $n = 2^a 3^b m$ where m has no factor 2 or 3, and compare the number of divisors of n with that of $2n$ as a ratio.
6. The odd divisors of n are exactly the divisors of the largest odd number dividing n. When does a number have an odd number of divisors?

Solutions

Problem 1

Answer: 377

If $n = p_1^{a_1} \cdots p_k^{a_k}$, then n has $(a_1+1)\cdots(a_k+1)$ divisors. For this product to equal 3, which is prime, there must be just one prime, with exponent 2, so $n = p^2$ for a prime p. The squares of primes below 200 are 4, 9, 25, 49, 121 and 169 (the next one, $17^2 = 289$, is too big). Their sum is 377.

Problem 2

Answer: 6

We have $7200 = 72 \times 100 = 2^5 \times 3^2 \times 5^2$, so a divisor is $2^a 3^b 5^c$ with $0 \le a \le 5$, $0 \le b \le 2$ and $0 \le c \le 2$. Being a multiple of 6 means $a \ge 1$ and $b \ge 1$. Being a multiple of 12 as well would mean $a \ge 2$ and $b \ge 1$. So the divisors we want have $b \ge 1$ and $a = 1$ exactly. Then a has 1 choice, b has 2 choices (1 or 2) and c has 3 choices, giving $1 \times 2 \times 3 = 6$ divisors.

Problem 3

Answer: 3

We have $12 = 2^2 \times 3$, $18 = 2 \times 3^2$ and $720 = 2^4 \times 3^2 \times 5$. Since n divides the LCM, $n = 2^a 3^b 5^c$ with $a \le 4$, $b \le 2$, $c \le 1$. The LCM takes the largest exponent of each prime. For the prime 2, the largest of $2, 1, a$ must be 4, which forces $a = 4$. For the prime 3, the largest of $1, 2, b$ must be 2, which holds for $b = 0, 1$ or 2. For the prime 5, the largest of $0, 0, c$ must be 1, which forces $c = 1$. So n is 80, 240 or 720, and there are 3 such n.

Problem 4

Answer: 9

Let $g = \text{hcf}(n^2+5, n+2)$. Since g divides $n+2$, it divides $(n+2)(n-2) = n^2-4$. It also divides n^2+5, so it divides the difference $(n^2+5)-(n^2-4) = 9$. Hence $g \le 9$ for every n. The value 9 is

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achieved: when $n = 7$ we get $n + 2 = 9$ and $n^2 + 5 = 54 = 6 \times 9$, whose HCF is 9. So the largest possible value is 9.

Problem 5

Answer: 72

Write $n = 2^a 3^b m$, where m is not divisible by 2 or 3. Then n has $(a+1)(b+1)d(m)$ divisors, $2n$ has $(a+2)(b+1)d(m)$, and $3n$ has $(a+1)(b+2)d(m)$, where $d(m)$ is the number of divisors of m. Dividing, $\frac{a+2}{a+1} = \frac{15}{12} = \frac{5}{4}$, so $4a + 8 = 5a + 5$ and $a = 3$. Similarly $\frac{b+2}{b+1} = \frac{16}{12} = \frac{4}{3}$, so $3b + 6 = 4b + 4$ and $b = 2$. Then $12 = 4 \times 3 \times d(m)$ gives $d(m) = 1$, so $m = 1$ and $n = 2^3 \times 3^2 = 72$. As a check, $144 = 2^4 \times 3^2$ has $5 \times 3 = 15$ divisors and $216 = 2^3 \times 3^3$ has $4 \times 4 = 16$.

Problem 6

Answer: 53

Write $n = 2^k m$ with m odd. The odd divisors of n are exactly the divisors of m, and a number has an odd number of divisors exactly when it is a perfect square. So we need $m = s^2$ for some odd s, that is, $n = 2^k s^2$.

If k is even, then $n = (2^{k/2}s)^2$ is a perfect square. If k is odd, then $n = 2 \times (2^{(k-1)/2}s)^2$ is twice a perfect square. Conversely, if n is a square or twice a square, every odd prime appears in n to an even power, so its odd part m is a square. Hence the numbers we want are exactly the squares and the doubles of squares. No number is both, since the exponent of 2 is even in a square and odd in twice a square.

There are 31 squares up to 1000 (as $31^2 = 961$ and $32^2 = 1024$), and the doubles of squares $2t^2 \le 1000$ need $t^2 \le 500$, so $t \le 22$ (as $22^2 = 484$ and $23^2 = 529$), giving 22 more. The answer is $31 + 22 = 53$.

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